Bangladesh Army University of Engineering & Technology

Department of Computer Science & Engineering
1st Year 2nd Semester
Course Code: CSE-1211
Course Title: Structured Programming

Class Test: 01 | **Full Marks: 15** | **Time: 30 Minutes**

Have you ever heard about Raya from the movie "Raya and the last dragon"? She will be your companion throughout this semester.

Marks

1. Raya wants to develop a system called "Area Finder". She wants to add the following features in her system: **05**

When using the system, a user is allowed to input an integer number among 5 and 7. If the user enters 5, then the system again asks for two inputs, height and base of a triangle from the user and calculates area of the triangle. If the user enters 7, then the system on the other hand, asks for one input, radius of a circle and calculates area of the circle. Then the system will print the area of the triangle or of the circle based on user's choice. In case of other inputs, other than 5 and 7, the system simply prints the message: "Invalid input" (not case sensitive).

As you are a CSE student, Raya seeks your help to implement the "Area Finder". Help her.

Sample Input 1	Sample Input 2
5 8 2	1
Sample Output 1	**Sample Output 2**
8	Invalid input

5 → height
→ base
7 → radius

*Use typecasting where necessary.

2. Raya had 4 dragons with whom she passed most of her time. Suddenly Raya thought, whether the health quality of all of her dragons is well or not. So, she wanted to find out the dragon with the lowest weight so that she can take care of that dragon well. Again, she seeks your help to do that. **05**

Your task is to develop a program that takes the weights of the 4 dragons and find the lowest weight.

Input: You will take 4 integer numbers (w1, w2, w3, w4).
Output: Print the lowest weight.

if (w1>
if (w1>w2)
else

Sample Input 1	Sample Input 2
1 2 3 4	4 3 2 1
Sample Output 1	**Sample Output 2**
1	1

3. Find the output of the following code snippet: **05**

```
#include<stdio.h>
int main(){
   int x = 50;
   if(x>50 && x<=50) printf("World\n");
   if((x!=50 && x>=50) || (x>=30 && x<=50)) printf("Hi\n");
   if(x>60 || x>70) printf("Hello\n");
   else printf("I konw Raya\n");
   if(2) printf("And she is wonderful\n");
}
```

highest

if(w1>w2)
printf("

*Maintain output format carefully

Scanned with CamScanner

Bangladesh Army University of Engineering & Technology

Department of Computer Science and Engineering

1st Year 2nd Semester (Summer-2024: 18th Batch)

Course Code & Title: CSE-1211 & Structured Programming

Class Test: 02 **Full Marks: 15** **Time: 25 Minutes**

1. Define Structured Programming Language. According to the rules of identifier naming in C, which of the following are valid identifiers, and which are invalid? Provide an explanation for your choices. [CO2, PO2, KP3, C2] 5

University of Engineering & Technology (BAUET)

studentAge	_tempValue	_userID	flag_status
max_value	firstName	static	1averageHeight
count123	isActive	total_Score	numItems

2. Demonstrate a flowchart to find the largest among the three numbers entered by the user. [CO2, PO2, KP3, C2] 5

3. Compare between Compiler and Interpreter. [CO2, PO2, KP3, C2] 5

Scanned with CamScanner

Bangladesh Army University of Engineering & Technology

Department of Computer Science & Engineering
1st Year 2nd Semester
Course Code: CSE-1211
Course Title: Structured Programming

Class Test: 02 | **Full Marks: 15** | **Time: 30 Minutes**

Remember Raya? → digit Split

Marks

1. Raya considers a number lucky if that number starts and ends with 7. For example, 7897 is a lucky number where 7889 is not. Your task is to develop a program that can detect whether a number is lucky or not according to Raya. **05**

2. Raya knows what an Armstrong number is. She knows that, an Armstrong number (also known as a narcissistic number) is a number that is equal to the sum of its own digits each raised to the power of the number of digits. For example, 1634 is an Armstrong number because, $1^4+6^4+3^4+4^4 = 1+1296+81+256 = 1634$. **05**

   Now write a program that can print the 25th Armstrong number; number starting from 1.

3. Write a code to print the following pattern: **05**

```
*****
 *****
  *****
   *****
    *****
```

while

sum = 0 + 7 n =

n = 1

for(n=1; n=

10|7897|789
70
89
80
97
90
7

while (n != 0)
n % 10 = 7

Scanned with CamScanner

Bangladesh Army University of Engineering & Technology

Department of Computer Science and Engineering

1st Year 2nd Semester (Summer-2024: 18th Batch)

Course Code & Title: CSE-1211 & Structured Programming

x, y

x, y

Class Test: 03 **Full Marks: 15** **Time: 25 Minutes**

1. Define the Scope and Lifetime of variables with proper examples. [CO2, PO2, KP3, C1] 3
2. Illustrate the advantages or disadvantages of using union over structure. [CO2, PO2, KP3, C2] 4
3. Demonstrate the **result** from the following expression. [CO2, PO2, KP3, C2] 4

   i. Assuming, m = 3, n = 1, and p = 2?
   int **result** = m << n + p * 1;

   Bodmas

   5

   ii. Assuming, x = 2, y = 3, and z = 5:
   int **result** = (x > y) && (y < z) || (z == 0);

   0

4. Difference between for, while, and do-while loop. [CO2, PO2, KP3, C4] 4

Scanned with CamScanner